

=====

256×256 ORIGINAL HASH — COMPLETE STATISTICAL ANALYSIS

Author: Marco Lazcano

=====

Input: 256 bytes
Output: 256 bytes
State: 256 bytes
Rounds: 24
Chunk: 256 bytes (full state, one pyramidal block)
Anchors: base -> tip (original order)
K1/K2: Sacred Pair boundary at dimension 129
K1: dimensions 0..127 (normal)
K2: dimensions 128..254 (sacred pair ACTIVE)
All tests on FULL 256-byte output. No shortcuts.

=====

=====

TEST 1: BASIC FUNCTIONALITY & SENSITIVITY

=====

Output: 256 bytes — first 32 hex chars shown

b" (empty)	-> df8b75b158179432ba480f2cb72d8eab...
b"\x00'	-> 752fedc0a8a797b79fab0306c20860f7...
b"\x01'	-> 81f677cac69682c8dbd214ae21a766e9...
b"\xff"	-> 4cfd8577dbc96149f414c5e65fb9f04b...
'Hola Mundo'	-> 0b4616015b393486be0569624ff05b51...
'Hello'	-> f4ddc5e54d7b7a05bb1f38efc6618743...
'hello'	-> ed223a7b9d04864bfd3b143f135ea863...
'hello ' (space)	-> 5ce94875f610487008839e36b487aa0c...

All hashes distinct: YES ✓

=====

TEST 2: DETERMINISM (200 trials)

=====

Errors: 0 / 200
Result: PASS ✓

=====

TEST 3: AVALANCHE EFFECT — FULL 2048-BIT OUTPUT (150 samples)

=====

Flipping 1 input bit — measuring all 2048 output bits

Mean: 49.8044% (ideal: 50.0000%)

Std Dev: 1.1960% (ideal: < 5%)
Minimum: 46.14%
Maximum: 52.73%
Result: EXCELLENT ✓

=====

TEST 4: SHANNON ENTROPY (300 samples, 256-byte output)

=====

Total bytes analyzed: 76,800
Unique byte values: 256 / 256
Entropy: 7.997477 bits/byte (maximum: 8.0)
Efficiency: 99.9685%
Result: EXCELLENT ✓

=====

TEST 5: CHI-SQUARE UNIFORMITY (600 samples)

=====

Total bytes: 153,600
Expected per value: 600.0
Chi-square: 268.77
Critical value 95%: 293.2 (255 degrees of freedom)
Critical value 99%: 310.4
Result: EXCELLENT ✓

=====

TEST 6: PEARSON CORRELATION — FULL 256-BYTE OUTPUT

=====

Samples: 1000 (N=1000 for statistical confidence)
Byte pairs: 32,640
Pairs/samples ratio: 32.6
Estimated time: ~25-30 minutes

Processed 100/1000... (184s elapsed, ~1659s remaining)
Processed 200/1000... (372s elapsed, ~1488s remaining)
Processed 300/1000... (550s elapsed, ~1284s remaining)
Processed 400/1000... (728s elapsed, ~1091s remaining)
Processed 500/1000... (904s elapsed, ~904s remaining)
Processed 600/1000... (1079s elapsed, ~720s remaining)
Processed 700/1000... (1258s elapsed, ~539s remaining)
Processed 800/1000... (1436s elapsed, ~359s remaining)
Processed 900/1000... (1615s elapsed, ~179s remaining)
Processed 1000/1000... (1793s elapsed, ~0s remaining)

Time: 1819.0 seconds

Pairs analyzed: 32,640
Average correlation: 0.02526 (ideal: ~0.000)
Maximum correlation: 0.13128
Pairs with corr > 0.10: 60 of 32,640 (0.18%)

NOTE: N=1000 provides statistical confidence to distinguish
sampling variance from structural correlation.
Average below 0.05 confirms genuine independence.
Result: EXCELLENT ✓

=====

TEST 7: S-BOX DIFFERENTIAL ANALYSIS

=====

Max differential probability: 0.0156 (1.56%)
Security threshold: 0.0625 (6.25%)
Result: SECURE ✓

=====

TEST 8: STRICT AVALANCHE CRITERION — SAC (150 samples)

=====

8 input bits x 2048 output bits — full 256-byte output
Estimated time: ~14 minutes on pure Python

Processed 25/150... (421s elapsed, ~2106s remaining)
Processed 50/150... (829s elapsed, ~1657s remaining)
Processed 75/150... (1233s elapsed, ~1233s remaining)
Processed 100/150... (1625s elapsed, ~812s remaining)
Processed 125/150... (2005s elapsed, ~401s remaining)
Processed 150/150... (2377s elapsed, ~0s remaining)

Time: 2376.7 seconds
Cells analyzed: 16,384
Cells with deviation >10%: 161 of 16384 (0.98%)
Result: EXCELLENT ✓

=====

TEST 9: COLLISION RESISTANCE (400 attempts)

=====

Processed 100/400...
Processed 200/400...
Processed 300/400...
Processed 400/400...
Time: 649.7 seconds
Collisions found: 0

Result: ROBUST ✓

=====

TEST 10: SACRED PAIR (K1 ↔ K2) VERIFICATION — DIM 129

=====

Double inversion sacred_pair(sacred_pair(x)) == x : PASS ✓
XOR property x XOR sacred_pair(x) == 0xFF : PASS ✓

Sample values:

x -> sacred_pair XOR

0 (0x00) -> 255 (0xff) 0xff
1 (0x01) -> 254 (0xfe) 0xff
127 (0x7f) -> 128 (0x80) 0xff
128 (0x80) -> 127 (0x7f) 0xff
254 (0xfe) -> 1 (0x01) 0xff
255 (0xff) -> 0 (0x00) 0xff

K1/K2 boundary in 256x256 Original Hash:

K1 = dimensions 0..127 -> normal anchor (unchanged)

K2 = dimensions 128..254 -> sacred_pair applied (ACTIVE)

Tip (dim 255) -> appended last

With 256-byte input: 255 anchors produced (i=0..254)

Sacred Pair ACTIVE for anchors i=128..254 = 127 anchors in K2

This differs from 512x512 Inverted Hash where K2 was never reached.

=====

TEST 11: PERFORMANCE (10 runs per size)

=====

Input	Avg ms	Note
256 bytes	1588.8 ms	prototype only
512 bytes	2203.4 ms	prototype only
1 KB	3211.8 ms	prototype only
4 KB	9796.4 ms	prototype only

Note: Pure Python — C/Rust implementation planned.

=====

FINAL REPORT — 256x256 ORIGINAL HASH

=====

256×256 ORIGINAL HASH — COMPLETE RESULTS

Test	Result	Status
Basic functionality	All distinct	✓
Determinism (200 trials)	0 errors	✓
Avalanche (2048 bits)	49.804% $\sigma=1.196\%$	✓
Shannon entropy	7.997477 b/B	✓
Chi-square uniformity	268.77	✓
Pearson corr. avg (N=1000)	0.02526	✓
Pearson corr. max	0.13128	✓
S-box max differential	0.0156 (1.56%)	✓
SAC deviation > 10%	0.98%	✓
Collision resistance	0 / 400 att.	✓
Sacred pair (K1↔K2)	Verified	✓

Input/Output/State: 256 bytes
Anchors: base->tip (original order, NOT inverted)
K1: dim 0..127 normal | K2: dim 128..254 SP active
Sacred Pair ACTIVE: 127 anchors in K2 region
Status: RESEARCH PROTOTYPE — Not for production

ANALYSIS COMPLETE — 256×256 ORIGINAL HASH

[Program finished]